Exact big Ramsey degree four for two-element subsets of the ordered set of integers

Joanna Boyland, William Gasarch, Nathan Hurtig, Robert Rust

arxiv:2305.07192v3; CC BY 4.0

Editorial note. The counted statement and its proof are adjacent in the source: the unlabelled proof environment beginning at source line 498 is the proof of Theorem the:Z2 printed immediately above it, and the article gives no other proof of this theorem.

Editorial note. The retained context includes the definition of the big Ramsey degree, which ends with a citation of Kechris, Pestov and Todorcevic at source line 227. The bibliography fragment retained for this source supplies that single entry together with the two further entries of the article, so the citation resolves and no reference is left undefined.

Editorial note. The original numbers theorems by section. The assembled wrapper keeps the author's own theorem environment, so the retained section opening from source line 465 keeps the numbering context of the original without inventing a new number.

Editorial note. Only the counted theorem is a counting unit. The n = 1 sibling and the general integers theorem T(n,zeta) = 2^n are not counted: the general theorem's printed lower bound is explicitly left to the reader at source lines 588-589, while the counted n = 2 theorem has a complete printed proof of both bounds.

Editorial scope. One counted exact value, the big Ramsey degree T(2,zeta) = 4 for two-element subsets of the ordered set of integers, with both bounds of the article's complete adjacent proof. The n = 1 sibling, the article's general integers theorem, the omega-squared example, the omega^d and omega^d times k theorems and the general theorem for ordinals below omega^omega are not counted.

Allowed standard prerequisites (editorial).

- Infinite Ramsey theorem, in the article's own big-Ramsey-degree form. For every n and every finite colouring of the n-element subsets of the naturals there is an infinite subset on which the colouring is constant; the article restates this as T(n, omega) = 1 at source lines 241-251.. source lines 239-251, retained as context. Applied twice in the counted proof, to the four-tuple colouring of the pairs of omega (source line 507) and, in the sibling arguments of the same section, to colourings of the naturals.
- Order equivalence and copies of the integers. Two ordered sets are order-equivalent when an order-preserving bijection exists between them; the

ordered set zeta is the integers with their natural order, and a subset of zeta is order-equivalent to zeta exactly when it has neither a least nor a greatest element and is unbounded in both directions.. definitions at source lines 172-196 and 213-216, retained as context. Both directions of the counted proof rebuild or test a subset H of zeta that is order-equivalent to zeta, and the lower bound uses that such an H must contain positive elements, negative elements and pairs of both signs.

- Elementary manipulations of Z with signs, absolute values and parity. Every integer is either a nonnegative integer or the negative of one; the absolute value of a negative integer is its negation; an infinite subset of the naturals can be split by the parity of its indices.. source lines 501-513 and 515-537. Building the four sign combinations of a pair, comparing absolute values in the four cases of the lower-bound colouring and alternating the signs of the sequence h_0, h_1, ... when the copy of zeta is rebuilt.

Definition 0.1. Let $\mathbb{N}$ be the set of natural numbers, including 0. Let $n, c \in \mathbb{N}$ and S be a set. Then we define $[c]$ to be $\{0, \dots, c-1\}$ (note $0 \in [c]$, but $c \notin [c]$). Note that if $c = 0$ then $[c] = \emptyset$. We use $\binom{S}{n}$ to denote the set of all n-element subsets of S.

Definition 0.2. A function $\mathrm{COL}\colon \binom{S}{n} \to [c]$ is called a *coloring*. A set $H \subseteq S$ is called *homogeneous* if COL restricted to $\binom{H}{n}$ is constant, that is, $\left|\mathrm{COL}\left(\binom{H}{n}\right)\right| \leq 1$.

Definition 0.3. Let $\mathcal{A} = (A, \preceq_A)$ and $\mathcal{B} = (B, \preceq_B)$ be ordered sets, where $\preceq_A, \preceq_B$ are total orders on A, B respectively.

1. Two ordered sets $\mathcal{A}, \mathcal{B}$ are *order-equivalent*, denoted $\mathcal{A} \approx \mathcal{B}$, if there exists an order-preserving bijection $f\colon A \to B$ such that, for all $a_1, a_2 \in A$:

$$a_1 \preceq_A a_2 \iff f(a_1) \preceq_B f(a_2).$$

2. The ordered set $\mathcal{A}+\mathcal{B}$ is defined to be $(A \sqcup B, \preceq)$, where the disjoint union $A \sqcup B$ is formally defined to be $(\{0\} \times A) \cup (\{1\} \times B)$, and where $\preceq$ is the dictionary ordering on $A \sqcup B$. Note that $+$ agrees with the definition of ordinal addition, and is not commutative in general.

3. Let $\mathcal{A} = (A, \preceq_A)$ be an ordered set with $A \subseteq \mathbb{Z}$. Then define $-\mathcal{A}$ to be $(-A, \preceq_{-A})$, where $-A = \{-a\colon a \in A\}$, and $-a \preceq_{-A} -b \iff b \preceq_A a$.

Definition 0.4. Let S be an ordered set, $S' \subseteq S$, and $n, c, t \in \mathbb{N}$. Let $\mathrm{COL}\colon \binom{S}{n} \to [c]$ be a coloring.

1. S' is *homogeneous* if $\left|\mathrm{COL}\left(\binom{S'}{n}\right)\right| = 1$.

2. S' is *t-homogeneous* if $\left|\mathrm{COL}\left(\binom{S'}{n}\right)\right| \leq t$.

3. S' is *S-t-homogeneous* if both S' is t-homogeneous and $S' \approx S$.

Definition 0.5. We identify every ordinal α as the ordered set of all ordinals $\beta < \alpha$, with 0 as the least ordinal. Additionally, let ω be the ordered set containing only the naturals, ζ be the ordered set containing only the integers, and η be the ordered set containing only the rationals, all under their respective natural orderings.

Definition 0.6. Let S be an ordered set. For any $n \in \mathbb{N}$, $T(n, S)$ is the least $t \in \mathbb{N}$ such that for all constants $c \geq 1$ and colorings $\mathrm{COL}\colon \binom{S}{n} \to [c]$, there exists some $S' \subseteq S$ such that S' is S-t-homogeneous. Note that t is independent of c. If no such t exists then we define $T(n, S) = \infty$. The quantity $T(n, S)$ is called the *big Ramsey degree* of n in S. The term was first coined by Kechris et al. [2]. Other sources use the Ramsey arrow notation modified for ordered sets. In that notation, $T(n, S) = t$ is short-hand for both $S \to (S)^n_{c,t}$ and $S \not\to (S)^n_{T(n,S),t-1}$ for all $c \in \mathbb{N}$. We use the $T(n, S)$ notation in this paper.

Ramsey's Theorem on $\mathbb{N}$ gives an infinite 1-homogeneous subset of $\mathbb{N}$. Theorem 0.7 restates Ramsey's Theorem using big Ramsey degree notation.

Theorem 0.7. *For all $n \in \mathbb{N}$, $T(n, \omega) = 1$.*

1 Big Ramsey Degrees of ζ

Our general result is $T(n, \zeta) = 2^n$. We first prove $T(1, \zeta) = 2$ and $T(2, \zeta) = 4$. Both of these will build notation and structure toward the main result.

Theorem 1.1. *The big Ramsey degree $T(2, \zeta) = 4$.*

Proof. We first prove $T(2, \zeta) \leq 4$. Let $c \in \mathbb{N}$ and let

$$\mathrm{COL}\colon \binom{\zeta}{2} \to [c]$$

be a c-coloring of $\binom{\zeta}{2}$. Let $\mathrm{COL}'\colon \binom{\omega}{2} \to [c]^4$ be defined by

$$\mathrm{COL}'(x, y) = (\mathrm{COL}(-x, -y), \mathrm{COL}(-x, y), \mathrm{COL}(x, -y), \mathrm{COL}(x, y))$$

where $x < y$.

By Theorem 0.7 there exists an ω-1-homogeneous set G. Let the color of the homogeneous set be (c_1, c_2, c_3, c_4). Let $G = \{h_0 < h_1 < \cdots\}$.

Then consider the set

$$H = \{-h_i\colon i \text{ is even}\} + \{h_i\colon i \text{ is odd}\},$$

which is order-equivalent to ζ. Let $s_i n_i, s_j n_j \in H$, where $n_i, n_j \geq 0$ and $s_i, s_j \in \{-1, 1\}$. Then by definition of COL', $\mathrm{COL}(s_i n_i, s_j n_j)$ is in $\{c_1, c_2, c_3, c_4\}$, depending on s_i and s_j. As an aside, this method only works when n_i and n_j are guaranteed to be distinct, which we forced with our alternation of sign by the parity of index. Since $\mathrm{COL}(H) \subseteq \{c_1, c_2, c_3, c_4\}$, H is ζ-4-homogeneous. Therefore $T(2, \zeta) \leq 4$.

We now prove $T(2,\zeta) \geq 4$. Let $\mathrm{COL}\colon \binom{\zeta}{2} \to [4]$ be the coloring

$$\mathrm{COL}(x,y) = \begin{cases} 0 & x,y \geq 0 \\ 1 & x \geq 0,\ y < 0, \text{ and } |x| \leq |y| \\ 2 & x \geq 0,\ y < 0, \text{ and } |x| > |y| \\ 3 & x < 0,\ y < 0. \end{cases}$$

Let H be an arbitrary order-equivalent subset of ζ. The following x, y are guaranteed to exist because $H \subseteq \zeta$ and $H \approx \zeta$:

1. Let $x, y \in H$ such that $x, y \geq 0$. Then $\mathrm{COL}(x,y) = 0$.
2. Let $x, y \in H$ such that $x \geq 0$, $y < 0$, and $|x| \leq |y|$. Then $\mathrm{COL}(x,y) = 1$.
3. Let $x, y \in H$ such that $x \geq 0$, $y < 0$, and $|x| > |y|$. Then $\mathrm{COL}(x,y) = 2$.
4. Let $x, y \in H$ such that $x < 0$, $y < 0$. Then $\mathrm{COL}(x,y) = 3$.

Therefore $T(2,\zeta) \geq 4$, and with the previous result, $T(2,\zeta) = 4$. □

References

[1] A. Blass, N. Dobrinen, and D. Raghavan. The next best thing to a P-point. *The Journal of Symbolic Logic*, 80(3):866–900, 2015.

[2] A. S. Kechris, V. G. Pestov, and S. Todorcevic. Fraïssé limits, Ramsey theory, and topological dynamics of automorphism groups. *Geometric & Functional Analysis GAFA*, 15(1):106–189, Feb 2005.

[3] F. Ramsey. On a problem of formal logic. *Proceedings of the London Math Society*, 30(1):264–286, 1930.